

NAME: Austin DeCiccoNET-ID: ajd295

1. Recall $\text{Exp}(\lambda)$ population has PDF: $f(x) = \frac{1}{\lambda} e^{-x/\lambda}, x > 0$.

The null hypothesis $\lambda = 10$ is to be tested against the alternative $\lambda = 5$ using observed sample data $x_1, x_2, \dots, x_n$. Use the Neyman Pearson Lemma, to obtain a test with the most power when the size of the critical region is to be a fixed α .

$$L_0 = f(x_1 | \lambda=10) f(x_2 | \lambda=10) \dots = \frac{1}{10^n} e^{-\frac{1}{10}(\sum x)} = \frac{1}{10^n} e^{-\frac{n\bar{x}}{10}}$$

$$L_1 = \frac{1}{5^n} e^{-\frac{n\bar{x}}{5}}$$

$$\frac{L_0}{L_1} = \frac{5^n}{10^n} (e^{n\bar{x}/5} e^{-n\bar{x}/10}) = \left(\frac{1}{2}\right)^n e^{\frac{n\bar{x}}{10}}$$

$$\frac{L_0}{L_1} \leq k$$

$$\left(\frac{1}{2}\right)^n e^{\frac{n\bar{x}}{10}} \leq k$$

$$e^{\frac{n\bar{x}}{10}} \leq k$$

$$\frac{n\bar{x}}{10} \leq k$$

$$\bar{x} \leq k$$

2. In the previous question suppose $n = 1$ (sample size of one) and the probability of type 1 error $\alpha = 0.05$. What is the critical region in this case?

$$n=1, \text{ thus } \bar{x} = x \quad \lambda=10 \quad \int_0^x \frac{1}{\lambda} e^{-x/\lambda} dx = 1 - e^{-x/\lambda} = F(x)$$

$$P(x \leq k) = \alpha$$

$$F(k) = \alpha = 1 - e^{-k/10}$$

$$1 - \alpha = e^{-k/10}$$

$$\alpha = 0.05$$

$$\ln(1 - \alpha) = -k/10$$

$$k = -10 \ln(1 - \alpha) = -10 \ln(0.95)$$